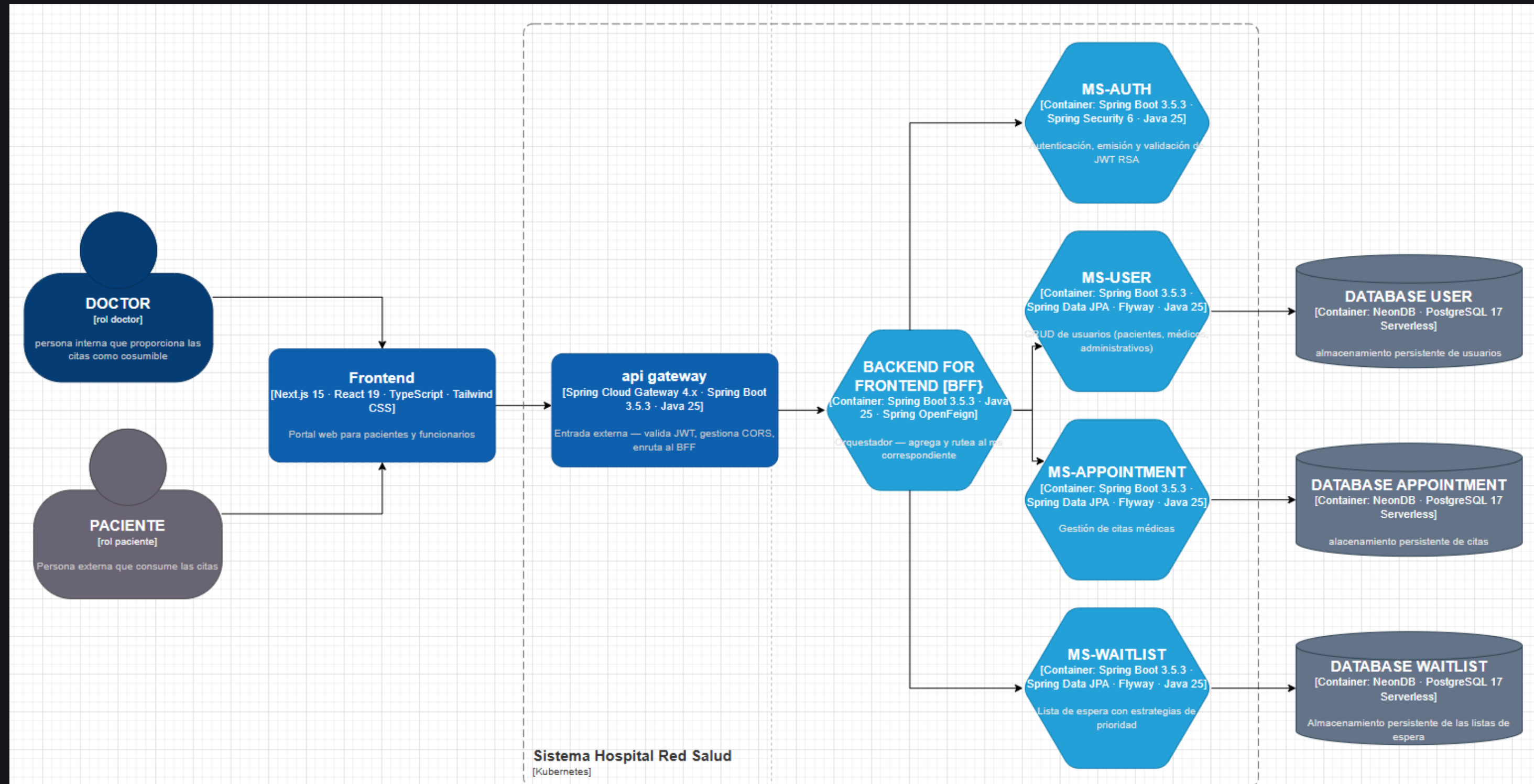


Sistema Gestión de Citas Médicas

DESARROLLO FULLSTACK III_001V

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Arquitectura de Microservicios



Flujo: Inicio de Sesión

1

Usuario ingresa credenciales

Email + contraseña en el Portal del Paciente
(o Funcionario) → localhost:3001

2

API Gateway valida JWT

Spring Cloud Gateway intercepta la request.
Rutas /api/auth/** son públicas → pasa sin token.

3

BFF llama a ms-auth

POST /api/auth/login → ms-auth
verifica BCrypt
y firma token JWT con clave
privada RSA 2048-bit.

4

Token retornado

JWT con rol
(PACIENTE/DOCTOR/ADMIN)
guardado
en sessionStorage. Redirige al
dashboard.

Seguridad JWT/RSA

Los demás microservicios validan
tokens descargando clave pública
desde /.well-known/jwks.json

ms-auth firma tokens con clave
privada RSA

Flujo: Toma de Citas



AppointmentStatus

PENDING

CONFIRMED

COMPLETED

CANCELED

NO_SHOW

Pruebas Unitarias

Backend

```
Chiza@DESKTOP-0UD6P7N MINGW64 ~/Desktop/HospitalSystem/backend/ms-appointment
$ ./gradlew test

BUILD SUCCESSFUL in 2s
5 actionable tasks: 5 up-to-date
Consider enabling configuration cache to speed up this build: https://docs.gradle.org/7.4.2/userguide/configuration_cache.html#sec:configuration_cache

Chiza@DESKTOP-0UD6P7N MINGW64 ~/Desktop/HospitalSystem/backend/ms-appointment
$ |
```

Test Summary

19 tests

0 failures

0 ignored

2.371s duration

100% successful

Packages

Classes

Package	Tests	Failures	Ignored	Duration	Success rate
com.hospital.mswaitlist.service	19	0	0	2.371s	100%

10 tests

0 failures

0 ignored

0.910s duration

100% successful

Tests

Standard output

Standard error

Test	Duration	Result
cancelByDoctor_notFound_returns404()	0.052s	passed
cancelByDoctor_returnsCancelledWithDoctorTag()	0.017s	passed
cancelByPatient_returnsCancelledWithPatientTag()	0.017s	passed
create_returnsCreated()	0.682s	passed
create_withInvalidBody_returnsBadRequest()	0.035s	passed
delete_returnsNoContent()	0.024s	passed
getAll_returnsOkWithList()	0.038s	passed
getById_found_returnsOk()	0.012s	passed
getById_notFound_returns404()	0.017s	passed
getByPatient_returnsOk()	0.016s	passed

JUnit 5

Mockito

Playwright

Frontend

```
Chiza@DESKTOP-0UD6P7N MINGW64 ~/Desktop/HospitalSystem/frontend
$ npm test

> frontend@0.1.0 test
> playwright test

Running 9 tests using 1 worker
9 passed (17.5s)

To open last HTML report run:
npx playwright show-report
```

login.spec.ts	
✓ Flujo de login › muestra error con credenciales inválidas chromium	8.1s
login.spec.ts:11	
✓ Flujo de login › bloquea a un doctor que intenta ingresar como paciente chromium	2.4s
login.spec.ts:30	
✓ Flujo de login › login exitoso como doctor redirige al dashboard chromium	1.4s
login.spec.ts:49	

muestra error con credenciales inválidas

login.spec.ts:11 8.1s

chromium

✓ Run

Test Steps

Q Filter steps

> ✓ Before Hooks 137ms

> ✓ Navigate to "/" — login.spec.ts:12 257ms

> ✓ Click getByText("Soy paciente") — login.spec.ts:15 101ms

> ✓ Fill "usuario@falso.cl" getByPlaceholder("Correo electrónico") — login.spec.ts:18 14ms

> ✓ Fill "ClaveIncorrecta123" getByPlaceholder("Contraseña") — login.spec.ts:19 9ms

> ✓ Click getByRole("button", { name: "Iniciar sesión" }) — login.spec.ts:22 33ms

> ✓ Expect "toBeVisible" getByText("Credenciales inválidas. Por favor, ingresa correo y contraseña válidos.") — login.spec.ts:27 7.5s

> ✓ After Hooks 24ms

Validación en kubernetes

```
Chiza@DESKTOP-OD6P7N MINGW64 ~/Desktop/HospitalSystem/frontend
$ kubectl get pods -n hospital
```

NAME	READY	STATUS	RES
bff-547fc86cd8-x5skv	1/1	Running	1 (
ms-appointment-69d55bc94f-7x6h4	1/1	Running	4 (
ms-appointment-69d55bc94f-9xt8j	1/1	Running	4 (
ms-appointment-postgres-868fc9ffdc-s9tf6	1/1	Running	1 (
ms-auth-779bd46f66-7wh64	1/1	Running	3 (
ms-auth-postgres-7dd55f6c68-9dw5x	1/1	Running	1 (
ms-user-86469745dd-jzs88	1/1	Running	3 (
ms-user-86469745dd-xscr4	1/1	Running	4 (
ms-user-postgres-6544c5b485-jftmw	1/1	Running	1 (
ms-waitlist-7fc699d4f8-p68jv	1/1	Running	4 (
ms-waitlist-7fc699d4f8-tskqf	1/1	Running	4 (

```
Chiza@DESKTOP-OD6P7N MINGW64 ~/Desktop/HospitalSystem/frontend
$ kubectl get services -n hospital
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP
bff-service	ClusterIP	10.96.30.186	<none>
ms-appointment-postgres-svc	ClusterIP	10.96.129.129	<none>
ms-appointment-service	ClusterIP	10.96.3.223	<none>
ms-auth-postgres-svc	ClusterIP	10.96.73.147	<none>
ms-auth-service	ClusterIP	10.96.24.213	<none>
ms-user-postgres-svc	ClusterIP	10.96.165.234	<none>
ms-user-service	ClusterIP	10.96.87.227	<none>
ms-waitlist-svc	ClusterIP	10.96.34.254	<none>

Todos los servicios
funcionan correctamente

./deploy.sh

Servicios aislados con
ClusterIP — sin
exposición externa

thank you